

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2023
Data Structures and Algorithms (CS-212T)
Assignment # 2

Semester: 3rd
Announced Date: 15-12-2023
Total Marks: 3
Instructor Name: Adeel Saeed

Batch: 2022F
Due Date: 22-12-2023
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 2	Apply different operations on linear data structures.	PLO_3 (Problem Analysis)	C3 (Apply)

Ques#1: Convert the following infix expression to its equivalent postfix expression showing stack contents for the conversion: A+B*(C-D)/E

Ques#2: Given a 5 elements stack S (from top to bottom: 2, 4, 6, 8, 10), and an empty queue Q, remove the elements one-by-one from S and insert them into Q, then remove them one-by-one from Q and re-insert them into S. Now, finally show the elements of S (from top to bottom).

Ques#3: Assume that you have an empty circular queue (array-based implementation) which can hold only four elements. What are the array indexes of the back and the front elements after executing this series of queue operations? (Indexes start with 0)

enqueue(“a”), enqueue(“b”), getFront(), enqueue(“c”), enqueue(“d”),
dequeue(), dequeue(), dequeue(), enqueue(“a”), enqueue(“b”),
enqueue(“c”), dequeue(), getFront(), enqueue(“d”), dequeue().